

DEEPAK SHARMA

Full-Stack Developer & Machine Learning Enthusiast

Bengaluru, India | +91-9050831620 | deepakshrma.work@gmail.com

[LinkedIn](#) • [GitHub](#)

I am a Computer Science undergraduate passionate about first-principles thinking, with a deep curiosity for Science, Mathematics, and Arts. Pursuing my degree through an industry-centric residential program, my expertise bridges software development, systems engineering, and applied machine learning. Currently I am playing around and learning how to build robust, scalable solutions from event-driven data pipelines to agentic AI workflows.

Technical Skills

Programming Languages: Python, JavaScript, Java, SQL, Data Structures & Algorithms

Machine Learning & AI: Neural Networks, CNNs, Object Detection, Linear/Logistic Regression, Decision Trees, Random Forests, KNN, DBSCAN, PCA, Anomaly Detection

Web & Backend: Node.js, Express, React.js, Mongoose, WebSockets/Stream SDK, JWT/OAuth, Worker queues (Inngest)

Databases: MongoDB, PostgreSQL, MySQL, Data modeling, Indexing

Libraries & Frameworks: PyTorch, scikit-learn

Experience & Leadership

Head of Operations | Entrepreneurship Cell (ECell), Scaler School of Technology

Bengaluru, India

Aug 2025 – June 2026

- Lead operations for campus events, managing logistics and cross-team coordination for high-impact summits, including the Mela Ventures Hackathon and masterclasses with industry leaders.
- Ensured the efficient execution of Pitchathon with a 2.5L prize pool and multiple such events.
- Collaborated with faculty mentors at the Scaler Innovation Lab to document strategic roadmaps and align execution with organizational goals.

Teaching Assistant | Scaler School of Technology

Bengaluru, India

Mar 2026 – June 2026

- Mentor a cohort of 23 junior students in programming, Data Structures & Algorithms, and web development through peer-learning and doubt-solving sessions.
- Provide academic and career guidance, fostering a growth mindset and supporting smooth onboarding into the curriculum.

Research & AI Projects

Inter-Sample Similarity Analysis in Text-to-Music Models

Research Project

- Conducted an empirical study on diversity and inter-sample similarity in text-to-music generation using the open-source HeartMula model.
- Generated and analyzed 1,760 audio clips to compare encoding-level interventions with prompt expansion strategies.
- Designed a qualitative evaluation framework to assess similarity and observed trade-offs between latent-space interventions and prompt engineering.

Corrective RAG System

Agentic AI, LLMs, Retrieval-Augmented Generation

- Implemented an Agentic AI Corrective RAG pipeline that validates retrieved context prior to response generation to reduce hallucinations.
- Built retrieval, reranking, and answer-generation workflows for grounded question answering, and deployed the application for interactive use.

Assistive Object Detection for Visually Impaired Users

Computer Vision, Deep Learning

- Explored and evaluated deep learning-based object detection pipelines aimed at detecting and localizing everyday objects in real-world scenes to assist visually impaired users.

Fake Social Media Account Detection

Machine Learning, Classification

- Developed a supervised learning classification pipeline utilizing profile-level behavioral features to detect fraudulent accounts.
- Performed feature engineering, compared classical machine learning models based on precision/recall trade-offs, and deployed via Streamlit.

Software Engineering Projects

Messaging and Collaboration Platform

React, Node.js, Inngest, Sentry

- Coded a messaging platform integrating Inngest for background jobs and Sentry for real-time error tracking, reducing production downtime.
- Secured user sessions with Clerk and enabled low-latency messaging via Stream.io.
- Translated a borrowed, provided design system into a responsive UI using Tailwind CSS, focusing on accurate, pixel-perfect code implementation.

Automated ETL Data Pipeline

Google Apps Script, GitHub Actions, Python

- Designed an event-driven pipeline triggering data validation via Google Apps Script immediately upon Google Sheet updates.
- Automated the Extract-Transform-Load workflow using GitHub Actions, eliminating data entry errors through strict type-checking and automated alerts.

Custom HTTP Server

Python, Socket Programming, TCP/IP

- Implemented a lightweight, multi-threaded HTTP server from first principles using Python and Socket Programming to understand TCP, HTTP parsing, and connection lifecycles.
- Engineered robust parsing logic to handle GET/POST methods and serve static files, optimizing throughput.

Energy Monitoring CLI Tool

Java, JDBC, MySQL

- Designed a normalized relational schema and applied indexing strategies in MySQL to speed up heavy aggregation queries for an electricity monitoring platform.
- Wrote Java JDBC functions to aggregate daily energy usage, billing estimates, and budgets into a wrapped CLI App.

Education

Bachelor of Science in Computer Science | BITS Pilani

Current CGPA: 8.3/10

Aug 2024 – Present

- Pursuing degree via the 4-year residential program at Scaler School of Technology.
- Curriculum emphasizes software development, applied computer science, machine learning, and systems.

Project Links

Inter-Sample Similarity Analysis in Text-to-Music Models

GitHub: <https://github.com/ChampDeepak/txt2music>

Paper: <https://github.com/ChampDeepak/txt2music/blob/main/paper/paper.pdf>

Corrective RAG System

GitHub: <https://github.com/ChampDeepak/simplerag>

Deployed: <https://simplerag.vercel.app/>

Assistive Object Detection for Visually Impaired Users

GitHub: https://github.com/ChampDeepak/Team_Panchtatv

Fake Social Media Account Detection

GitHub: <https://github.com/ChampDeepak/FakeAccDetection>

Deployed: <https://fakeaccddetection.streamlit.app/>

Messaging and Collaboration Platform

GitHub: <https://github.com/ChampDeepak/slack-clone>

Deployed: <https://slack-clone-frontend-fawn.vercel.app/>

Automated ETL Data Pipeline

GitHub: <https://github.com/ChampDeepak/etlPipeline>

Custom HTTP Server

GitHub: <https://github.com/ChampDeepak/httpbasedserver>